Worksheet 2.6 Factorizing Algebraic Expressions

Section 1 Finding Factors

Factorizing algebraic expressions is a way of turning a sum of terms into a product of smaller ones. The product is a multiplication of the factors. Sometimes it helps to look at a simpler case before venturing into the abstract. The number 48 may be written as a product in a number of different ways:

$$48 = 3 \times 16 = 4 \times 12 = 2 \times 24$$

So too can polynomials, unless of course the polynomial has no factors (in the way that the number 23 has no factors). For example:

$$x^3 - 6x^2 + 12x - 8 = (x-2)^3 = (x-2)(x-2)(x-2) = (x-2)(x^2 - 4x + 4)$$

where $(x-2)^3$ is in fully factored form.

Occasionally we can start by taking common factors out of every term in the sum. For example,

$$\begin{aligned} 3xy + 9xy^2 + 6x^2y &= 3xy(1) + 3xy(3y) + 3xy(2x) \\ &= 3xy(1 + 3y + 2x) \end{aligned}$$

Sometimes not all the terms in an expression have a common factor but you may still be able to do some factoring.

<u>Example 1</u> :

$$9a^2b + 3a^2 + 5b + 5b^2a = 3a^2(3b+1) + 5b(1+ba)$$

<u>Example 2</u> :

$$\begin{aligned} 10x^2 + 5x + 2xy + y &= 5x(2x+1) + y(2x+1) \qquad \text{Let } T = 2x+1 \\ &= 5xT + yT \\ &= T(5x+y) \\ &= (2x+1)(5x+y) \end{aligned}$$

<u>Example 3</u> :

$$\begin{aligned} x^2 + 2xy + 5x^3 + 10x^2y &= x(x+2y) + 5x^2(x+2y) \\ &= (x+5x^2)(x+2y) \\ &= x(1+5x)(x+2y) \end{aligned}$$

Exercises:

1. Factorize the following algebraic expressions:

   (a) $6x + 24 = 3(2x+8)$

   (b) $8x^2 - 4x = 4x(2x-1)$

   (c) $6xy + 10x^2y = 2xy(3+10x)$

   (d) $m^4 - 3m^2 = m^2(m^2-3)$

   (e) $6x^2 + 8x + 12yx = 2x(3x+4+6y)$

   For the following expressions, factorize the first pair, then the second pair:

   (f) $8m^2 - 12m + 10m - 15 = 4m(2m-3) + 5(2m-3)$

   (g) $x^2 + 5x + 2x + 10 = x(x+5) + 2(x+5)$

   (h) $m^2 - 4m + 3m - 12 = m(m+4) + 3(m-4)$

   (i) $2t^2 - 4t + t - 2 =$

   (j) $6y^2 - 15y + 4y - 10$

Section 2 Some standard factorizations

Recall the distributive laws of section 1.10.

<u>Example 1</u> :

$$\begin{aligned}(x+3)(x-3) &= x(x-3) + 3(x-3)\\ &= x^2 - 3x + 3x - 9\\ &= x^2 - 9\\ &= x^2 - 3^2\end{aligned}$$

<u>Example 2</u> :

$$\begin{aligned}(x+9)(x-9) &= x(x-9) + 9(x-9)\\ &= x^2 - 9x + 9x - 81\\ &= x^2 - 81\\ &= x^2 - 9^2\end{aligned}$$

Notice that in each of these examples, we end up with a quantity in the form $A^2 - B^2$. In example 1, we have

$$\begin{aligned} A^2 - B^2 &= x^2 - 9 \\ &= (x+3)(x-3) \end{aligned}$$

where we have identified $A = x$ and $B = 3$. In example 2, we have

$$\begin{aligned} A^2 - B^2 &= x^2 - 81 \\ &= (x+9)(x-9) \end{aligned}$$

where we have identified $A = x$ and $B = 9$. The result that we have developed and have used in two examples is called the difference of two squares, and is written:

$$A^2 - B^2 = (A+B)(A-B)$$

The next common factorization that is important is called a perfect square. Notice that

$$\begin{aligned} (x+5)^2 &= (x+5)(x+5) \\ &= x(x+5) + 5(x+5) \\ &= x^2 + 5x + 5x + 25 \\ &= x^2 + 10x + 25 \\ &= x^2 + 2(5x) + 5^2 \end{aligned}$$

The perfect square is written as:

$$(x+a)^2 = x^2 + 2ax + a^2$$

Similarly,

$$\begin{aligned} (x-a)^2 &= (x-a)(x-a) \\ &= x(x-a) - a(x-a) \\ &= x^2 - ax - ax + a^2 \\ &= x^2 - 2ax + a^2 \end{aligned}$$

For example,

$$\begin{aligned} (x-7)^2 &= (x-7)(x-7) \\ &= x(x-7) - 7(x-7) \\ &= x^2 - 7x - 7x + 7^2 \\ &= x^2 - 14x + 49 \end{aligned}$$

Exercises:

1. Expand the following, and collect like terms:

   (a) $(x+2)(x-2)$
   (b) $(y+5)(y-5)$
   (c) $(y-6)(y+6)$
   (d) $(x+7)(x-7)$
   (e) $(2x+1)(2x-1)$
   (f) $(3m+4)(3m-4)$
   (g) $(3y+5)(3y-5)$
   (h) $(2t+7)(2t-7)$

2. Factorize the following:

   (a) x^2-16
   (b) y^2-49
   (c) x^2-25
   (d) $4x^2-25$
   (e) $16-y^2$
   (f) m^2-36
   (g) $4m^2-49$
   (h) $9m^2-16$

3. Expand the following and collect like terms:

   (a) $(x+5)(x+5)$
   (b) $(x+9)(x+9)$
   (c) $(y-2)(y-2)$
   (d) $(m-3)(m-3)$
   (e) $(2m+5)(2m+5)$
   (f) $(t+10)(t+10)$
   (g) $(y+8)^2$
   (h) $(t+6)^2$

4. Factorize the following:

   (a) y^2-6y+9
   (b) $x^2-10x+25$
   (c) $x^2+8x+16$
   (d) $x^2+20x+100$
   (e) $m^2+16m+64$
   (f) $t^2-30t+225$
   (g) $m^2-12m+36$
   (h) $t^2+18t+81$

Section 3 Introduction to Quadratics

In the expression $5t^2 + 2t + 1$, t is called the variable. Quadratics are algebraic expressions of one variable, and they have degree two. Having degree two means that the highest power of the variable that occurs is a squared term. The general form for a quadratic is

$$ax^2 + bx + c$$

Note that we assume that a is not zero because if it were zero, we would have $bx + c$ which is not a quadratic: the highest power of x would not be two, but one. There are a few points to make about the quadratic $ax^2 + bx + c$:

1. a is the coefficient of the squared term and $a \neq 0$.
2. b is the coefficient of x and can be any number.
3. c is the called the constant term (even though a and b are also constant), and can be any number.

Quadratics may factor into two linear factors:

$$ax^2 + bx + c = a(x + k)(x + l)$$

where $(x + k)$ and $(x + l)$ are called the linear factors.

Exercises:

1. Which of the following algebraic expressions is a quadratic?

   (a) $x^2 - 3x + 4$
   (b) $4x^2 + 6x - 1$
   (c) $x^3 - 6x + 2$
   (d) $\frac{1}{x^2} + 2x + 1$
   (e) $x^2 - 4$
   (f) $6x^2$

Section 4 Factorizing Quadratics

Before we start factorizing quadratics, it would be a good idea to look for a pattern.

$$\begin{aligned}(x+2)(x+4) &= x^2+4x+2x+8 \\ &= x^2+6x+8\end{aligned}$$

Notice that the numbers 2 and 4 add to give 6 and multiply to give 8.

$$\begin{aligned}(x+5)(x-3) &= x^2-3x+5x-15 \\ &= x^2+2x-15\end{aligned}$$

Notice that the numbers 5 and -3 add to give 2 and multiply to give -15.

Let's try to factorize expressions similar to those above, where we will start with the expression in its expanded out form. To factorize the expression $x^2+7x+12$, we will try to find numbers that multiply to give 12 and add to give 7. The numbers that we come up with are 3 and 4, so we write

$$x^2+7x+12=(x+3)(x+4)$$

This equation should be verified by expanding the right hand side.

Example 1 : Factorize $x^2+9x+14$.

We attempt to find two numbers that add to give 9 and multiply to give 14, and the numbers that do this are 2 and 7. Therefore

$$x^2+9x+14=(x+2)(x+7)$$

Again, this equation shouldn't be believed until the right hand side is expanded, and is shown to equal $x^2+9x+14$.

Example 2 : Factorize $x^2+7x-18$.

We attempt to find two numbers that add to give 7 and multiply to give -18 (notice the minus!). The numbers that do this are -2 and 9. Therefore

$$x^2+7x-18=(x-2)(x+9)$$

This equation shouldn't be believed until the right hand side is expanded, and is shown to equal $x^2+7x-18$.

Exercises:

1. Factorize the following quadratics:

   (a) $x^2 + 4x + 3$
   (b) $x^2 + 15x + 44$
   (c) $x^2 + 11x - 26$
   (d) $x^2 + 7x - 30$
   (e) $x^2 + 10x + 24$
   (f) $x^2 - 14x + 24$
   (g) $x^2 - 7x + 10$
   (h) $x^2 - 5x - 24$
   (i) $x^2 + 2x - 15$
   (j) $x^2 - 2x - 15$

The method that we have just described to factorize quadratics will work, if at all, only in the case that the coefficient of x^2 is 1. For other cases, we will need to factorize by

1. Using the 'ACE' method, or by
2. Using the quadratic formula

The 'ACE' method (pronounced a-c), unlike some other methods, is clear and easy to follow, as each step leads logically to the next. If you can expand an expression like $(3x + 4)(2x - 3)$, then you will be able to follow this technique.

Example	Factorize $6x^2 - x - 12$
1: Multiply the first term $6x^2$ by the last term (-12) 2: Find factors of $-72x^2$ which add to $-x$. 3: Return to the original expression and replace $-x$ with $-9x + 8x$. 4: Factorize $(6x^2 - 9x)$ and $(8x - 12)$. 5: One common factor is $(2x - 3)$. The other factor, $(3x + 4)$, is found by dividing each term by $(2x - 3)$.	$-72x^2$ $(-9x)(8x) = -72x^2$ $-9x + 8x = -x$ $6x^2 - x - 12$ $= 6x^2 - 9x + 8x - 12$ $= 3x(2x - 3) + 4(2x - 3)$ $= (2x - 3)(3x + 4)$
6: Verify the factorization by expansion	$(3x + 4)(2x - 3)$ $= 3x(2x - 3) + 4(2x - 3)$ $= 6x^2 - 9x + 8x - 12$ $= 6x^2 - x - 12$

<u>Example 3</u> : Factorize $4x^2 + 21x + 5$.

1. Multiply first and last terms: $4x^2 \times 5 = 20x^2$
2. Find factors of $20x^2$ which add to $21x$ and multiply to give $20x^2$. These are $20x$ and x.
3. Replace $21x$ in the original expression with $20x + x$:

$$4x^2 + 21x + 5 = 4x^2 + 20x + x + 5$$

4. Factorize the first two terms and the last two terms

$$4x^2 + 20x + x + 5 = 4x(x + 5) + (x + 5)$$

5. Factorize further:

$$4x(x+5)+(x+5)=(x+5)(4x+1)$$

Exercises:

1. Factorize the following quadratics using the 'ACE' method:

(a) $2x^2+11x+12$
(b) $3x^2+16x+5$
(c) $6x^2+17x+12$
(d) $2x^2+9x+10$
(e) $12x^2+11x+2$
(f) $2x^2-5x-3$
(g) $3x^2-10x-8$
(h) $3x^2-11x-20$
(i) $5x^2+17x+6$
(j) $10x^2+19x+6$

Section 5 The quadratic formula

When there is no obvious whole-number solution to the quadratic factorization, the quadratic formula must be used. It can be shown by the method of completing the square that the solutions to $ax^2+bx+c=0$ are given by

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

If we let the roots be k and l, say, then

$$\begin{aligned} k &= \frac{-b+\sqrt{b^2-4ac}}{2a} \\ l &= \frac{-b-\sqrt{b^2-4ac}}{2a} \end{aligned}$$

Then

$$ax^2+bx+c=a(x-k)(x-l)$$

When factorizing using this method be sure to multiply throughout by the coefficient of x^2.

Example 1 : Factorize x^2+5x+3. We try the 'ACE' method. The obvious factors of $3x^2$ are $3x$ and $1x$, which won't add to $5x$. We abandon this method, and go to

the quadratic formula. Note that $a = 1$, $b = 5$, and $c = 3$.

$$\begin{aligned} x &= \frac{-5 \pm \sqrt{5^2 - 4(1)(3)}}{2(1)} \\ &= -\frac{5}{2} \pm \frac{\sqrt{13}}{2} \end{aligned}$$

so that the two roots are

$$k_1 = \frac{-5 + \sqrt{13}}{2} \quad \text{and} \quad k_2 = \frac{-5 - \sqrt{13}}{2}$$

Then

$$x^2 + 5x + 3 = (x - \frac{-5 + \sqrt{13}}{2})(x - \frac{-5 - \sqrt{13}}{2})$$

<u>Example 2</u> : Factorize $2x^2 - x - 5$.

Note that $a = 2$, $b = -1$, and $c = -5$. Then the solutions to $2x^2 - x - 5 = 0$ are

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{1 \pm \sqrt{(-1)^2 - 4 \times 2 \times (-5)}}{2 \times 2} \\ &= \frac{1 \pm \sqrt{41}}{4} \end{aligned}$$

So the two factors of $2x^2 - x - 5$ are

$$(x - \frac{1 + \sqrt{41}}{4}) \qquad \text{and} \qquad (x - \frac{1 - \sqrt{41}}{4})$$

and so the factorization is

$$2x^2 - x - 5 = 2\left(x - \frac{1 + \sqrt{41}}{4}\right)\left(x - \frac{1 - \sqrt{41}}{4}\right)$$

This right hand side of this equation should be expanded before it is believed!

Exercises:

1. Factorize the following quadratics using the quadratic formula:

   (a) $3x^2 + 2x - 4$
   (b) $x^2 + 3x + 1$
   (c) $2x^2 + 8x + 3$
   (d) $3x^2 + 5x + 1$
   (e) $3x^2 + 6x + 2$
   (f) $5x^2 + 7x - 2$
   (g) $3x^2 + 5x - 4$
   (h) $2x^2 + 4x + 1$
   (i) $5x^2 + 2x - 2$
   (j) $2x^2 + x - 7$

Section 6 Uses of factorization

We can use factorization of expressions in a variety of ways. One way is to simplify algebraic fractions.

Example 1 :

$$\begin{aligned}\frac{x^2-9}{x-3} &= \frac{(x-3)(x+3)}{(x-3)} \\ &= \frac{x-3}{x-3} \times (x+3) \\ &= x+3\end{aligned}$$

Example 2 :

$$\begin{aligned}\frac{x}{x^2+4x+4} + \frac{x}{x+2} &= \frac{x}{(x+2)^2} + \frac{x}{x+2} \\ &= \frac{x}{(x+2)^2} + \frac{x}{x+2} \times \frac{x+2}{x+2} \\ &= \frac{x}{(x+2)^2} + \frac{x^2+2x}{(x+2)^2} \\ &= \frac{x^2+x+2x}{(x+2)^2} \\ &= \frac{x(x+3)}{(x+2)^2}\end{aligned}$$

Another way of using factorization is in solving quadratic equations.

Example 3 : Solve $(x+3)^2 = x+5$.

$$\begin{aligned} x^2+6x+9 &= x+5 \\ x^2+5x+4 &= 0 \\ (x+4)(x+1) &= 0 \end{aligned}$$

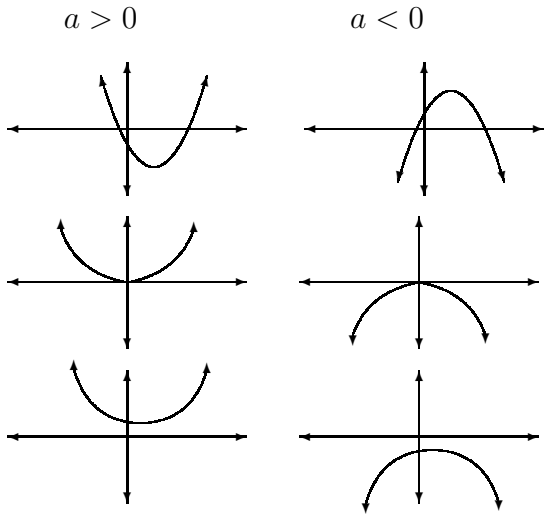
This is true when either $x=-4$ or $x=-1$. In other words, just one of the factors needs to be zero for the original equation that we started with to be true.

It is a good idea to know what to expect from the equation by first examining the discriminant $\Delta = b^2 - 4ac$. This is the expression under the square-root sign in the quadratic formula. Given the equation $y = ax^2 + bx + c$, and using our knowledge of square roots, we find the following:

$\Delta > 0$ There will be 2 distinct solutions, so the curve crosses the x-axis twice.

$\Delta = 0$ There will only be one solution, so the curve will only touch the x-axis. This is called a double root.

$\Delta < 0$ The curve does not touch the x-axis. We will deal with this case in detail later.



Exercises:

1. Factorize and then simplify the following algebraic expressions:

   (a) $\frac{x^2+3x}{x+3}$

   (b) $\frac{6x^2-8}{2x}$

   (c) $\frac{x^2+3x+2}{3x+6}$

   (d) $\frac{x^2-7x-18}{x^2-6x-27}$

   (e) $\frac{x^2-16}{2x+8}$

   (f) $\frac{3x^2-9x}{18x}$

(g) $\frac{x^2-25}{x^2-3x-10}$

(h) $\frac{2x^2-32}{x^2+6x+8}$

(i) $\frac{x^3-9x^2}{3x-27}$

(j) $\frac{2x^2-x-6}{x^2+x-6}$

2. Simplify the following by first finding a common denominator:

(a) $\frac{3}{x+2} + \frac{5x}{x+3}$

(b) $\frac{4x}{x-5} - \frac{2}{x+2}$

(c) $\frac{x+1}{x+2} + \frac{x+3}{x+4}$

(d) $\frac{6}{x^2+5x+6} + \frac{2}{x^2+8x+15}$

(e) $\frac{4}{x^2-3x-10} - \frac{1}{x^2+5x+6}$

(f) $\frac{x+3}{x^2+6x+9} - \frac{2}{x+3}$

(g) $\frac{x^2+8x+15}{x^2+7x+10} - \frac{x+3}{x+2}$

(h) $\frac{x^2-9}{2x+6} - \frac{x^2}{x-3}$

(i) $\frac{x}{x+1} - \frac{2x}{x+3}$

(j) $\frac{3x}{x^2+6x} - \frac{2x+1}{x+6}$

3. Solve the following quadratic equations:

(a) $x^2 - 6x + 8 = 0$

(b) $x^2 + 8x + 15 = 0$

(c) $x^2 + 7x + 12 = 0$

(d) $x^2 + 9x - 22 = 0$

(e) $x^2 - 7x + 12 = 0$

(f) $2x^2 - x - 6 = 0$

(g) $2x^2 - 13x - 7 = 0$

(h) $3x^2 - 10x - 8 = 0$

(i) $7x^2 + 13x - 2 = 0$

(j) $x^2 - 18x + 77 = 0$

4. Solve the following equations using the quadratic formula. Write the answers to two decimal places.

(a) $x^2 - 3x + 1 = 0$

(b) $2x^2 - 6x - 7 = 0$

(c) $3x^2 + 2x - 2 = 0$

(d) $2x^2 - 13x + 7 = 0$

Section 7 Multiplication and Division of Algebraic Fractions

We are often able to use factorization when we are multiplying or dividing algebraic expressions.

Example 1 :

$$\begin{aligned}\frac{x^2-16}{x+3}\times\frac{x^2+5x+6}{x+4} &= \frac{(x+4)(x-4)}{x+3}\times\frac{(x+3)(x+2)}{x+4}\\ &= (x-4)(x+2)\end{aligned}$$

Example 2 :

$$\begin{aligned}\frac{2x^2+12x+16}{3x^2+6x}\times\frac{4x^2-100}{6x+30} &= \frac{2(x^2+6x+8)}{3x(x+2)}\times\frac{4(x^2-25)}{6(x+5)}\\ &= \frac{2(x+4)(x+2)}{3x(x+2)}\times\frac{4(x+5)(x-5)}{6(x+5)}\\ &= \frac{4(x+4)(x-5)}{9x}\end{aligned}$$

Example 3 :

$$\begin{aligned}\frac{6x^2+9x}{x^2+8x+15}\div\frac{4x+6}{x^2-9} &= \frac{6x^2+9x}{x^2+8x+15}\times\frac{x^2-9}{4x+6}\\ &= \frac{3x(2x+3)}{(x+3)(x+5)}\times\frac{(x+3)(x-3)}{2(2x+3)}\\ &= \frac{3x(x-3)}{2(x+5)}\end{aligned}$$

Exercises:

1. Simplify the following expressions:
   (a) $\frac{2x^2-5x-3}{x^2+2x}\times\frac{x^2+4x}{2x+1}$
   (b) $\frac{3x+21}{x^2-7x+12}\times\frac{4x-12}{9x+63}$
   (c) $\frac{x^2+2x}{x^2+x-20}\times\frac{2x^2-5x-12}{5x+10}$
   (d) $\frac{3x^2-10x-8}{5x-15}\div\frac{2x^2-7x-4}{x^2-3x}$
   (e) $\frac{x^2-16}{4x^2-1}\div\frac{x^2+11x+28}{2x^2+5x+2}$

Exercises 2.6 Factorizing Algebraic Expressions

1. Expand

   (a) $(x+2)(x-2)$

   (b) $(2x+4y)(2x-4y)$

2. Factorize

   (a) $144x^2 - y^2$

   (b) $16a^2 - 9b^2$

   (c) $x^2 + 7x + 10$

   (d) $b^2 + 9b + 14$

   (e) $x^2 + 14x + 49$

   (f) $x^2 + x - 6$

   (g) $2x^3 + 10x^2 - 48x$

   (h) $2x^2 + 7x + 3$

   (i) $5b^2 + 17b + 6$

   (j) $18y^2 + 12y + 2$

   (k) $12x^2 + 6x - 6$

   (l) $7a^2 - 9a - 10$

   (m) $-x^2 - x + 2$

   (n) $-2x^2 + 3x + 2$

3. (a) Factorize $x^2 + 5x + 3$ using the quadratic formula.

   (b) Factorize $2x^2 - x - 1$.

   (c) Simplify $\frac{A^2-4}{3A-6}$.

   (d) Simplify $\frac{2x^2-8}{2x^2-x-6}$.

   (e) Simplify $\frac{x^2-x-6}{2xy} \times \frac{2x^2y}{x^2-9}$.

   (f) Simplify $\frac{2x^2+5x-3}{x^3+3x^2+2x} \div \frac{4x^2-1}{x^3+2x^2}$.

   (g) Simplify $\frac{3x}{x^2+6x+9} + \frac{x+3}{x^2-9}$.

   (h) Solve $x^2 + 2x - 3 = 0$.

Answers 2.6

Section 1

1. (a) $6(x+4)$
   (b) $4x(2x-1)$
   (c) $2xy(3+5x)$
   (d) $m^2(m^2-3)$
   (e) $2x(3x+4+6y)$
   (f) $(4m+5)(2m-3)$
   (g) $(x+5)(x+2)$
   (h) $(m-4)(m+3)$
   (i) $(t-2)(2t+1)$
   (j) $(2y-5)(3y+2)$

Section 2

1. (a) x^2-4
   (b) y^2-25
   (c) y^2-36
   (d) x^2-49
   (e) $4x^2-1$
   (f) $9m^2-16$
   (g) $9y^2-25$
   (h) $4t^2-49$
2. (a) $(x+4)(x-4)$
   (b) $(y+7)(y-7)$
   (c) $(x+5)(x-5)$
   (d) $(2x+5)(2x-5)$
   (e) $(4+y)(4-y)$
   (f) $(m+6)(m-6)$
   (g) $(2m+7)(2m-7)$
   (h) $(3m+4)(3m-4)$
3. (a) $x^2+10x+25$
   (b) $x^2+18x+81$
   (c) y^2-4y+4
   (d) m^2-6m+9
   (e) $4m^2+20m+25$
   (f) $t^2+20t+100$
   (g) $y^2+16y+64$
   (h) $t^2+12t+36$
4. (a) $(y-3)^2$
   (b) $(x-5)^2$
   (c) $(x+4)^2$
   (d) $(x+10)^2$
   (e) $(m+8)^2$
   (f) $(t-15)^2$
   (g) $(m-6)^2$
   (h) $(t+9)^2$

Section 3

1. (a), (b), (e), and (f)

Section 4 part 1

1. (a) $(x+3)(x+1)$
   (b) $(x+11)(x+4)$
   (c) $(x+13)(x-2)$
   (d) $(x+10)(x-3)$
   (e) $(x+6)(x+4)$
   (f) $(x-12)(x-2)$
   (g) $(x-5)(x-2)$

(h) $(x-8)(x+3)$
(i) $(x+5)(x-3)$
(j) $(x-5)(x+3)$

Section 4 part 2

1. (a) $(2x+3)(x+4)$
   (b) $(3x+1)(x+5)$
   (c) $(2x+3)(3x+4)$
   (d) $(2x+5)(x+2)$
   (e) $(3x+2)(4x+1)$
   (f) $(2x+1)(x-3)$
   (g) $(3x+2)(x-4)$
   (h) $(3x+4)(x-5)$
   (i) $(5x+2)(x+3)$
   (j) $(5x+2)(2x+3)$

Section 5

1. (a) $3(x-\frac{-2+\sqrt{52}}{6})(x-\frac{-2-\sqrt{52}}{6})$
   (b) $(x-\frac{-3+\sqrt{5}}{2})(x-\frac{-3-\sqrt{5}}{2})$
   (c) $2(x-\frac{-8+\sqrt{40}}{4})(x-\frac{-8-\sqrt{40}}{4})$
   (d) $3(x-\frac{-5+\sqrt{13}}{6})(x-\frac{-5-\sqrt{13}}{6})$
   (e) $3(x-\frac{-6+\sqrt{12}}{6})(x-\frac{-6-\sqrt{12}}{6})$
   (f) $5(x-\frac{-7+\sqrt{89}}{10})(x-\frac{-7-\sqrt{89}}{10})$
   (g) $3(x-\frac{-5+\sqrt{73}}{6})(x-\frac{-5-\sqrt{73}}{6})$
   (h) $2(x-\frac{-4+\sqrt{8}}{4})(x-\frac{-4-\sqrt{8}}{4})$
   (i) $5(x-\frac{-2+\sqrt{44}}{10})(x-\frac{-2-\sqrt{44}}{10})$
   (j) $2(x-\frac{-1+\sqrt{57}}{4})(x-\frac{-1-\sqrt{57}}{4})$

Section 6

1. (a) x
   (b) $\frac{3x^2-4}{x}$
   (c) $\frac{x+1}{3}$
   (d) $\frac{x+2}{x+3}$
   (e) $\frac{x-4}{2}$
   (f) $\frac{x-3}{6}$
   (g) $\frac{x+5}{x+2}$
   (h) $\frac{2(x-4)}{x+2}$
   (i) $\frac{x^2}{3}$
   (j) $\frac{2x+3}{x+3}$

2. (a) $\frac{5x^2+13x+9}{(x+2)(x+3)}$
   (b) $\frac{2(2x^2+3x+5)}{(x+2)(x-5)}$
   (c) $\frac{2(x^2+5x+5)}{(x+2)(x+4)}$
   (d) $\frac{2(4x+17)}{(x+2)(x+3)(x+5)}$
   (e) $\frac{3x+17}{(x+2)(x+3)(x-5)}$
   (f) $\frac{-1}{x+3}$
   (g) 0
   (h) $\frac{-x^2-6x+9}{2(x-3)}$
   (i) $\frac{-x(x-1)}{(x+1)(x+3)}$
   (j) $\frac{-2(x-1)}{(x+6)}$

3. (a) $4,\ 2$
   (b) $-5,\ -3$
   (c) $-4,\ -3$
   (d) $-11,\ 2$
   (e) $4,\ 3$
   (f) $-2,\ \frac{3}{2}$
   (g) $7,\ -\frac{1}{2}$
   (h) $4,\ -\frac{2}{3}$
   (i) $-2,\ \frac{1}{7}$
   (j) $11,\ 7$

4. (a) $\frac{3\pm\sqrt{5}}{2}$
   (b) $\frac{3}{2}\pm\frac{\sqrt{92}}{4}$
   (c) $-\frac{1}{3}\pm\frac{\sqrt{28}}{6}$
   (d) $\frac{13}{4}\pm\frac{\sqrt{113}}{4}$

Section 7

1. (a) $\frac{(x+4)(x-3)}{x+2}$
   (b) $\frac{4}{3(x-4)}$
   (c) $\frac{x(2x+3)}{5(x+5)}$
   (d) $\frac{x(3x+2)}{5(2x+1)}$
   (e) $\frac{(x-4)(x+2)}{(2x-1)(x+7)}$

Exercises 2.6

1. (a) $x^2 - 4$
   (b) $4x^2 - 16y^2$

2. (a) $(12x - y)(12x + y)$
   (b) $(4a - 3b)(4a + 3b)$
   (c) $(x + 5)(x + 2)$
   (d) $(b + 7)(b + 2)$
   (e) $(x + 7)^2$
   (f) $(x + 3)(x - 2)$
   (g) $2x(x - 3)(x + 8)$
   (h) $(x + 3)(2x + 1)$
   (i) $(b + 3)(5b + 2)$
   (j) $2(3y + 1)^2$
   (k) $6(x + 1)(2x - 1)$
   (l) $(7a + 5)(a - 2)$
   (m) $(x + 2)(1 - x)$
   (n) $(2x + 1)(2 - x)$

3. (a) $[x - (\frac{-5-\sqrt{13}}{2})][x - (\frac{-5+\sqrt{13}}{2})]$ or $(\frac{2x+5+\sqrt{13}}{2})(\frac{2x+5-\sqrt{13}}{2})$ or $\frac{1}{4}(2x + 5 + \sqrt{13})(2x + 5 - \sqrt{13})$
   (b) $(2x + 1)(x - 1)$
   (c) $\frac{A+2}{3}$
   (d) $\frac{2(x+2)}{2x+3}$
   (e) $\frac{x(x+2)}{x+3}$
   (f) $\frac{x(x+3)}{(2x+1)(x+1)}$
   (g) $\frac{4x^2-3x+9}{(x+3)^2(x-3)}$
   (h) $x = -3$ or $x = 1$